Software Engineering

BCT(III/I)



COCOMO

COCOMO

Overview on COCOMO

- Constructive Cost estimation Model (COCOMO)
- Proposed by Boehm in 1981
- Heuristic Project Estimation Technique
- Uses both Single and Multivariable estimation

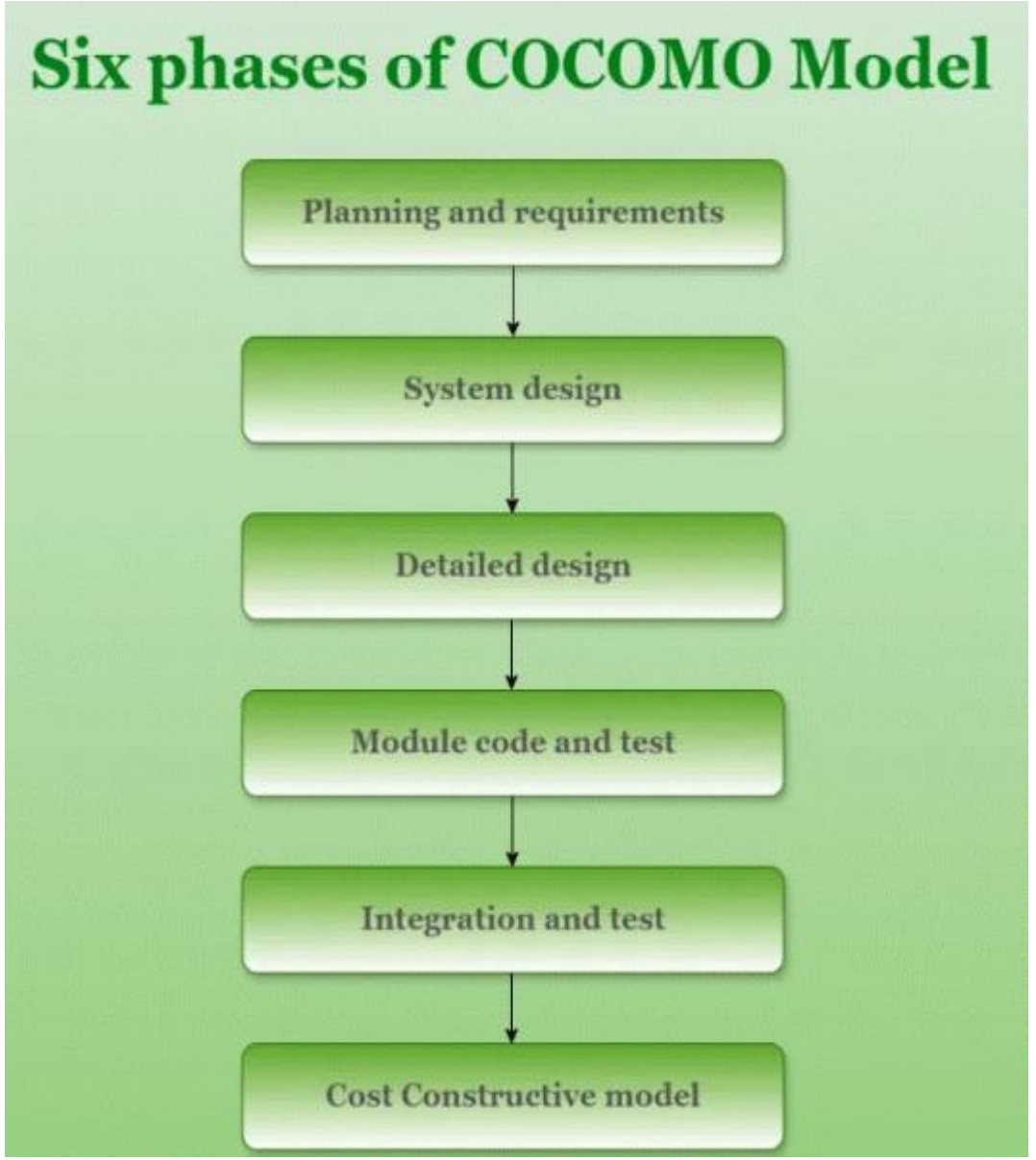
COCOMO

1. Planning and Requirements

- ❑ Define what the project will do, its goals, and any limits.
- ❑ Make a project plan showing the timeline, needed resources, and important checkpoints.

2. System Design

- ❑ Create the overall structure of the software.
- ❑ Decide the main parts, how they connect, and how data moves between them.



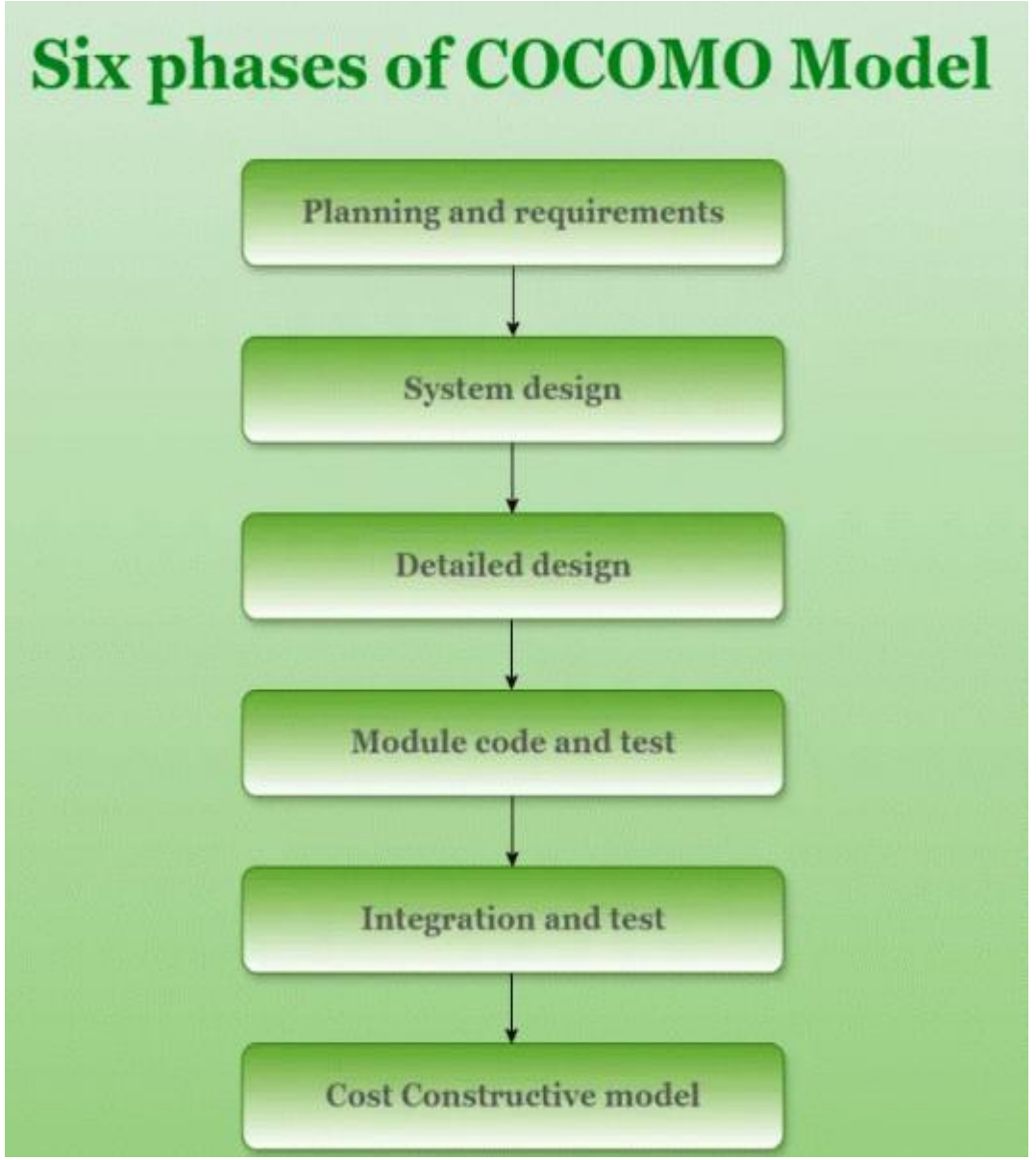
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3. Detailed Design

- ❑ Describe each part of the system in detail.
- ❑ Include the data structures, algorithms, and how parts will connect interface.

4. Module Code and Test

- ❑ Write the actual code for each part as planned in the detailed design.
- ❑ Build the functions, logic, and connections.



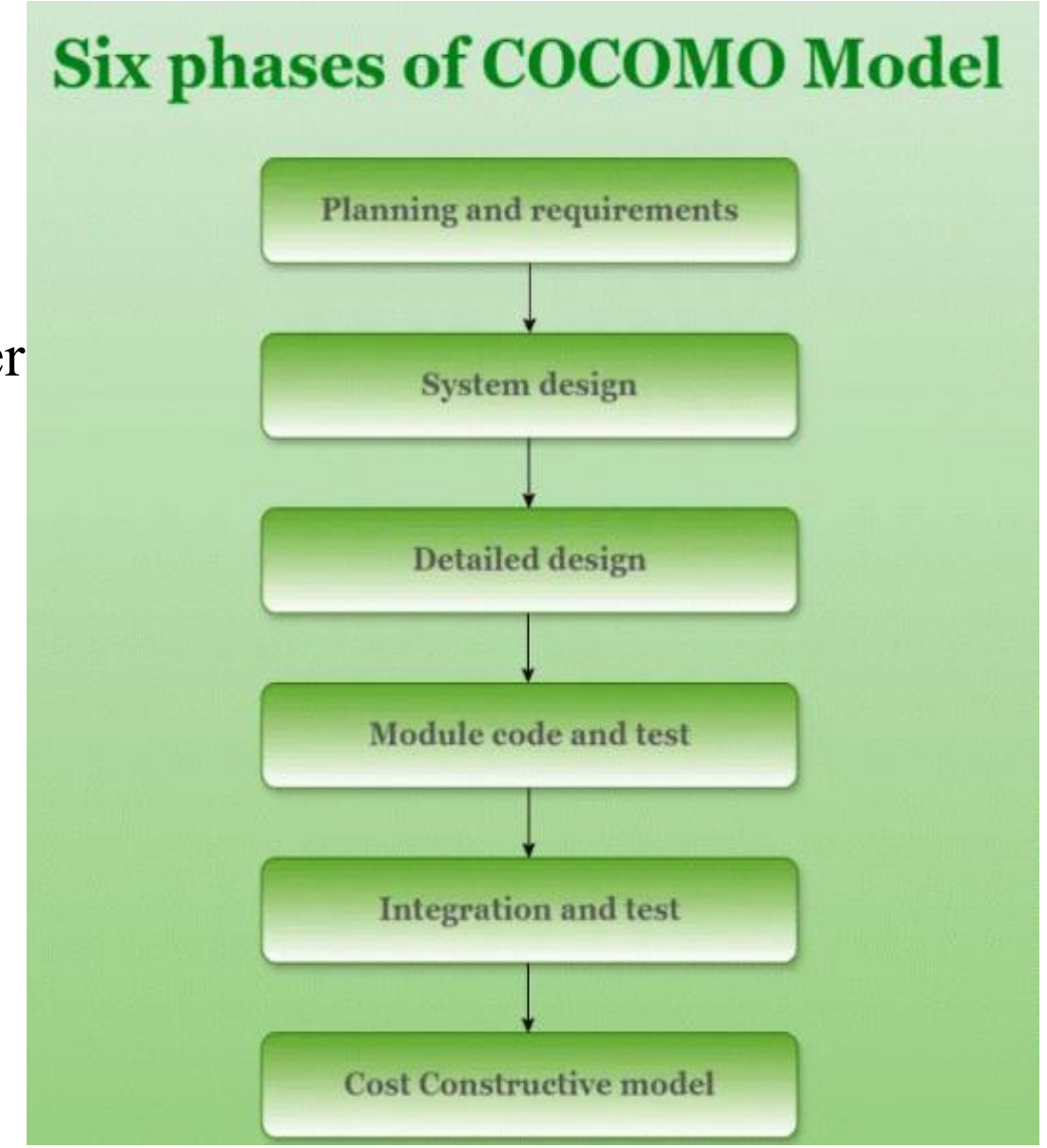
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5. Integration and Test

- ❑ Combine all parts to form the full system.
- ❑ Test to make sure everything works together properly.

6. Cost Constructive Model (COCOMO)

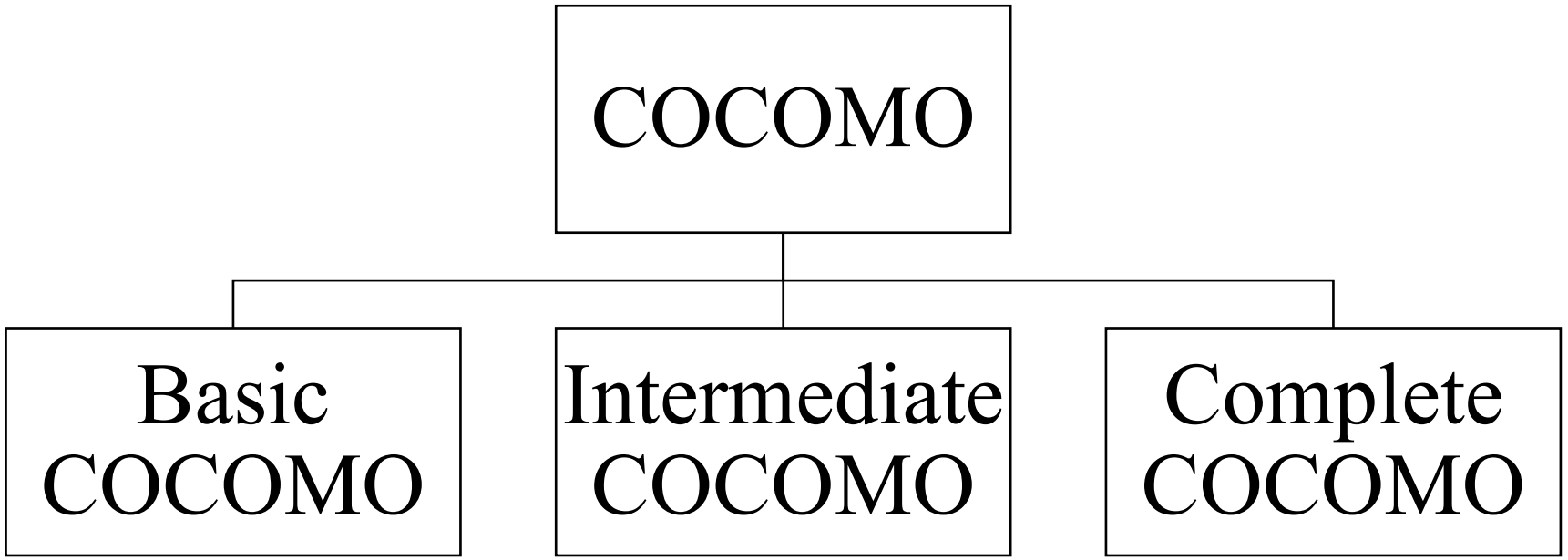
- ❑ A commonly used method to estimate how much time, money, and effort the software project will need.



COCOMO

Stages of COCOMO

- According to Boehm, COCOMO prescribes a three stage process estimation:



COCOMO

Basic COCOMO

- Used for quick and slightly rough calculations of Software Costs.
- Its accuracy is somewhat restricted due to the absence of sufficient factor considerations.
- Gives only an approximate estimation:
  - Effort = $a^* (KLOC)^b$ PM
  - $T_{dev} = c^* (Effort)^d$ months

COCOMO

Basic COCOMO

- KLOC is the estimated Kilo Lines of Source Code.
- a, b, c, d are the constants for different categories of software products.
- T_{dev} is the estimated time to develop the software products
- Effort estimation is obtained in terms of person-month (PMs)

COCOMO

- Types of Software Development Project
- According to Boehm, it is again classified into three categories based on Development complexity-
  - Organic
  - Semidetached
  - Embedded

COCOMO

Software Projects	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

COCOMO

Basic COCOMO

❑ Organic Complexity:

- Project should be well-defined application program
- Size of the development team is reasonably small
- Team was experienced in developing similar type of projects

COCOMO

Basic COCOMO

- Organic Complexity:
  - Formula for effort estimation :
    - Effort = 2.4* $(KLOC)^{1.05}$ PM
  - Formula for development time estimation :
    - T_{dev} = 2.5* $(Effort)^{0.38}$ months

COCOMO

Basic COCOMO

- Semidetached Complexity:
  - Development team is comprised of both experienced and in-experienced members
  - Team may have limited experience in developing similar type of project and may be unfamiliar with some aspects of the system.

COCOMO

Basic COCOMO

- Semidetached Complexity:
  - Formula for effort estimation :
    - $E = 3.0* (KLOC)^{1.12}$ PM
  - Formula for development time estimation :
    - $T_{dev} = 2.5* (Effort)^{0.35}$ months

COCOMO

Basic COCOMO

❑ Embedded Complexity:

- Developed software is highly coupled to hardware.
- Stringent regulations on the operational procedures exist.
- Team may have limited experience on related systems and may be unfamiliar with some aspects of the system.

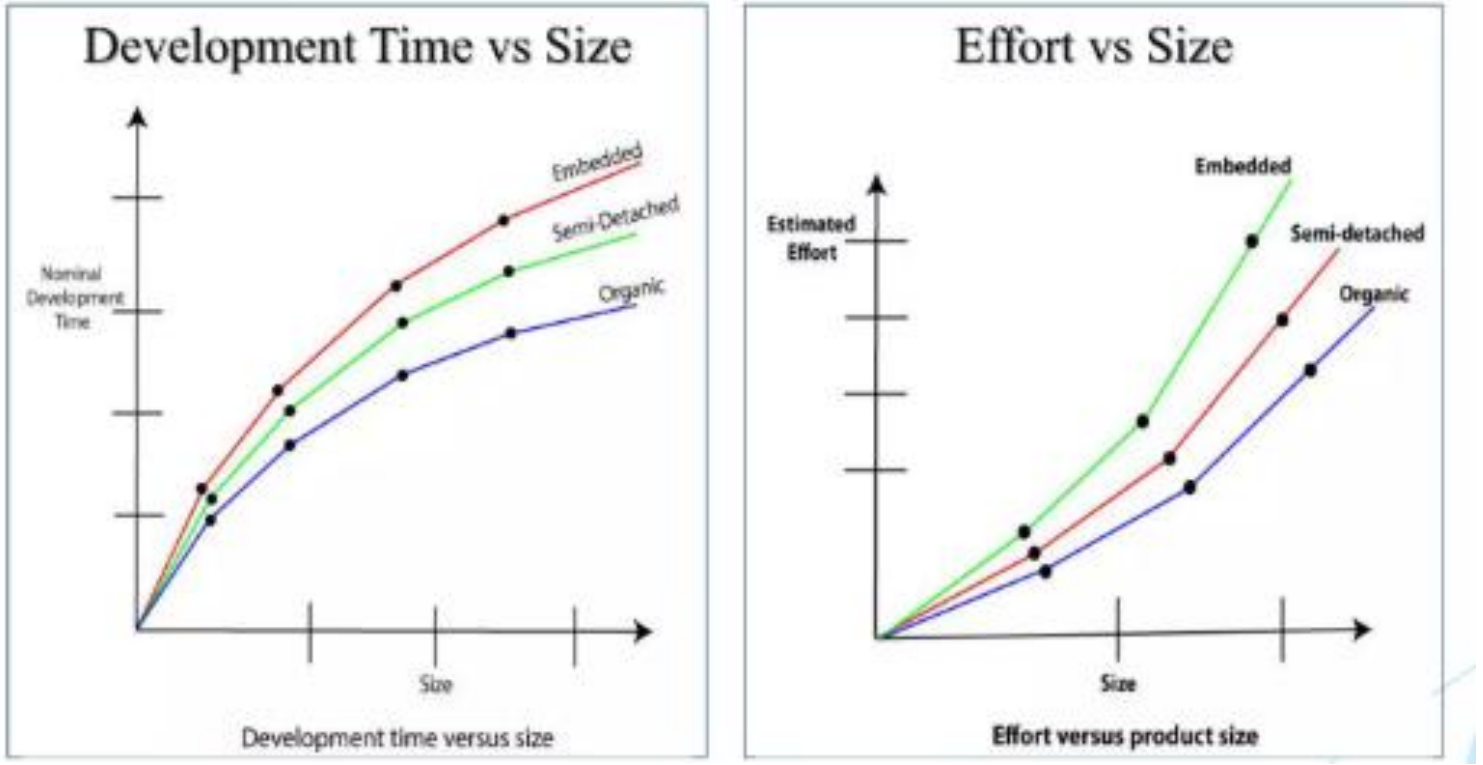
COCOMO

Basic COCOMO

- Embedded Complexity:
  - Formula for effort estimation :
    - Effort = 3.6* $(KLOC)^{1.20}$ PM
  - Formula for development time estimation :
    - T_{dev} = 2.5* $(Effort)^{0.32}$months

COCOMO

Graphical Representation



COCOMO

Basic COCOMO

Example: Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded.

Solution: The basic COCOMO equation takes the form:

Estimated Size of project= 400 KLOC

(i) Organic Mode

$E = 2.4 * (400)^{1.05} = 1295.31$ PM

$T = 2.5 * (1295.31)^{038} = 38.07$ months

COCOMO

Basic COCOMO

Example: Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded.

Solution:

The basic COCOMO equation takes the form:

Estimated Size of project= 400 KLOC

(ii) Semidetached Mode

$E = 3.0 * (400)^{1.12} = 2462.79$ PM

$T = 2.5 * (2462.79)^{035} = 38.45$ months

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Basic COCOMO

Example: Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded.

Solution:

The basic COCOMO equation takes the form:

Estimated Size of project= 400 KLOC

(iii) Embedded Mode

$E = 3.6 * (400)^{1.20} = 4772.81$ PM

$T = 2.5 * (4772.8)^{0.32} = 38$ months

COCOMO

Intermediate COCOMOI

- It refines the initial estimate obtained by using
- the Basic COCOMO.
- Uses 15 cost drivers that are multiplied with the
- initial cost and effort estimated by Basic
- COCOMO.
- As per Boehm, the scale of cost drivers is in
- between O and 3.

COCOMO

Intermediate COCOMO

- The cost drivers can be classified as being attributes of :
  - Product
  - Computer
  - Personnel
  - Development Environment

COCOMO

Intermediate COCOMO

- Product
  - Required software reliability extent
  - Size of the application database
  - The complexity of the product
- Computer
  - Run-time performance constraints
  - Memory constraints
  - The volatility of the virtual machine environment
  - Required turn-about time

COCOMO

Intermediate COCOMO

- ❑ Development Environment
  - Use of software tools
  - Application of software engineering methods
  - Required development schedule
- ❑ Personnel
  - Analyst capability
  - Software engineering capability
  - Applications experience
  - Virtual machine experience
  - Programming language experience

COCOMO

Complete COCOMO

- Incorporates all qualities of the standard version with an assessment of the cost drivers.
- Consider software product as a bunch of sub-systems.
- The development team might have no experience on similar type of projects.
- Development cost is estimated on the sub-systems.

COCOMO

Complete COCOMO

- Six phases Of complete COCOMO are:
  - Planning and requirements
  - System structure
  - Complete structure
  - Module code and test
  - Integration and test
  - Cost Constructive model

COCOMO-II

COCOMO 11

- COCOMO II incorporates a range of sub-models that produce detailed software estimate.
- Provide accurate cost and schedule estimates.
- Provide careful, easy-to-understand definitions of the Model's input, output and assumptions.
- Provides a constructive, normative and evolving model.

COCOMO-II

COCOMO 11

- There are four sub-models of COCOMO 11.
  - **Application Composition Model:** When software is composed from existing parts.
  - **Early Design Model:** When requirements are available but design has not yet started.
  - **Reuse Model:** Used to compute the effort of integrating reusable components.
  - **Post-architecture Model:** When system architecture is ready and more information is available.

Thank You!